PAPER_947: GW190425 Mass-Gap Phonon Classification

Author: Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-12 **Session:** 213 **Source:** ns_phonon_gw190425_wstp.py (MassGapPhononClassifier) **Calculator:** GW190425MassGapPhononCalc (CP4 #531) **CVW:** v2.0.0 compliant

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Abstract

We apply UQFF SCm suppression threshold classification to the heavier component of GW190425, measured at $m_1 = 2.52\,M_\odot$. This mass lies at the boundary between neutron stars and black holes ($M_{\text{boundary}} = 2.5\,M_\odot$). Using a sigmoid-based classifier with $\sigma = 0.1\,M_\odot$, we obtain $P(\text{NS}) = 49\%$ and $P(\text{BH}) = 51\%$, reflecting genuine physical ambiguity at the mass gap.

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1. Classification Formula

$$P(\text{BH}) = \frac{1}{1 + \exp\left(-\frac{m_1 - M_{\text{boundary}}}{\sigma}\right)}$$

$$P(\text{NS}) = 1 - P(\text{BH})$$

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2. GW190425 Parameters

Parameter	Value	Source
m_1	$2.52\,M_\odot$	LIGO/Virgo O3a
m_2	$1.27\,M_\odot$	LIGO/Virgo O3a
M_{boundary}	$2.5\,M_\odot$	UQFF SCm threshold
σ	$0.1\,M_\odot$	Classification width

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3. Classification Results

m_1 ($M_\odot$)	$P(\text{NS})$	$P(\text{BH})$	Classification
2.00	99.3%	0.7%	NS
2.30	88.1%	11.9%	NS
2.50	50.0%	50.0%	Boundary
2.52	45.0%	55.0%	BH (marginal)
2.70	11.9%	88.1%	BH
3.00	0.7%	99.3%	BH

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4. Physical Interpretation

The SCm suppression threshold corresponds to the mass at which internal phonon modes become damped by gravitational compression beyond neutron degeneracy. At $m_1 = 2.52\,M_\odot$, the system sits $0.02\,M_\odot$ above the boundary, yielding near-equal probabilities – consistent with LIGO/Virgo's classification uncertainty.

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5. Source Data

- **File:** ns_phonon_gw190425_wstp.py
- **Session:** 213
- **CP4 Class:** GW190425MassGapPhononCalc (#531)

References

1. Murphy, D.T. – Star Magic UQFF Framework (2024-2026)
2. Abbott, B.P. et al. (2020) – ApJL, 892, L3 (GW190425)
3. Tauris, T.M. et al. (2017) – ApJ, 846, 170